

```
In [20]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.svm import SVC
```

```
In [21]: df = pd.read_csv("car_acceptability.csv")
df
```

```
Out[21]:
```

	buying price	maintenance cost	number of doors	number of persons	lug_boot	safety	evaluation
0	vhigh	vhigh	2	2	small	med	unacc
1	vhigh	vhigh	2	2	small	high	unacc
2	vhigh	vhigh	2	2	med	low	unacc
3	vhigh	vhigh	2	2	med	med	unacc
4	vhigh	vhigh	2	2	med	high	unacc
...	...	...	...	...	...	...	...
1722	low	low	5	5	med	med	good
1723	low	low	5	5	med	high	vgood
1724	low	low	5	5	big	low	unacc
1725	low	low	5	5	big	med	good
1726	low	low	5	5	big	high	vgood

1727 rows × 7 columns

```
In [22]: le = LabelEncoder()

df['buying price'] = le.fit_transform(df['buying price'])
```

```
df['maintenance cost'] =le.fit_transform(df['maintenance cost'])
df['lug_boot']=le.fit_transform(df['lug_boot'])
df['safety']=le.fit_transform(df['safety'])
df['evaluation']=le.fit_transform(df['evaluation'])
```

```
In [23]: X=df.drop("evaluation",axis=1)
Y=df['evaluation']
```

```
In [24]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

```
In [25]: svm_model = SVC(kernel='linear')
svm_model.fit(X_train,Y_train)
```

```
Out[25]:
```

▼ SVC ⓘ ?

- Parameters
- Fitted attributes

```
In [26]: Y_pred = svm_model.predict(X_test)
```

```
In [27]: print("Accuracy:", accuracy_score(Y_test, Y_pred))
cm = confusion_matrix(Y_test, Y_pred)
print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))
```

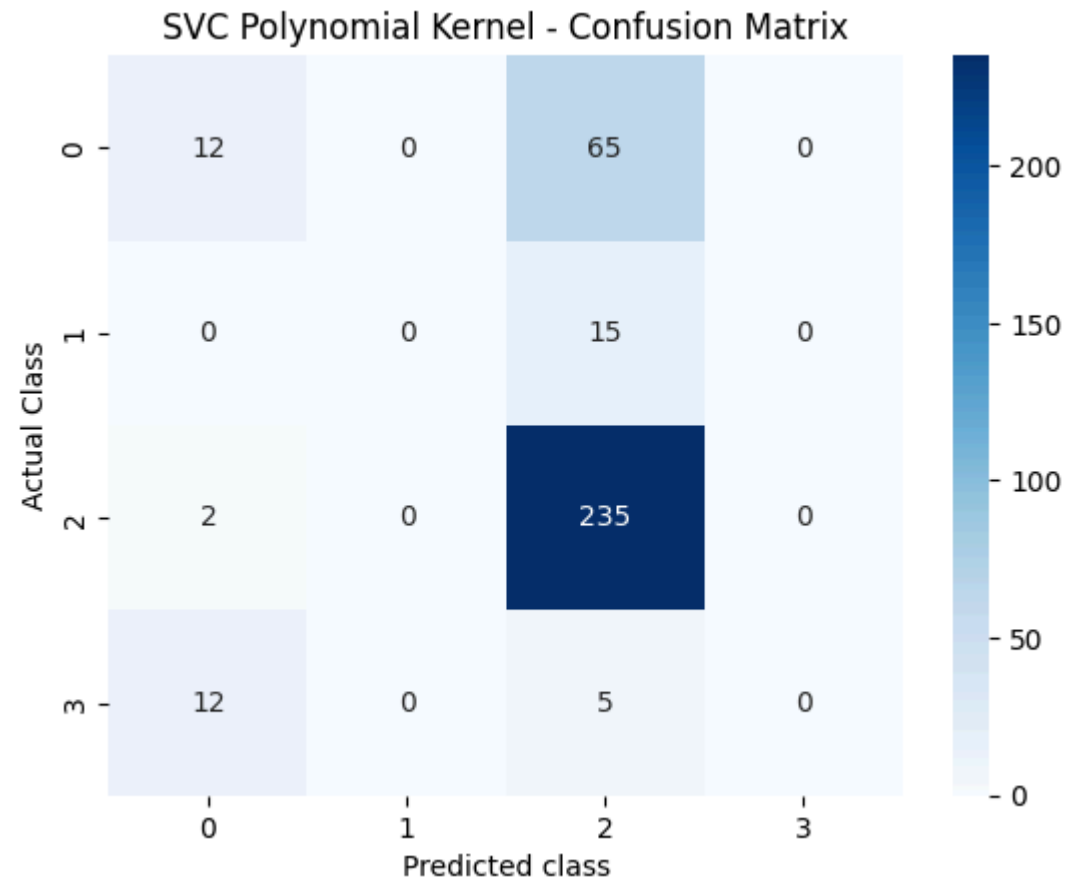
Accuracy: 0.7138728323699421

Confusion Matrix:

```
[[ 12   0  65   0]
 [  0   0  15   0]
 [  2   0 235   0]
 [ 12   0   5   0]]
```

```
In [28]: sns.heatmap(cm,annot=True,fmt="d",cmap="Blues")
plt.xlabel("Predicted class")
plt.ylabel("Actual Class")
```

```
plt.title("SVC Polynomial Kernel - Confusion Matrix")  
plt.show()
```



```
In [29]: svm_model = SVC(kernel='poly')  
svm_model.fit(X_train,Y_train)
```

Out[29]:

▼ SVC ⓘ ?

► Parameters

► Fitted attributes

In [30]: `Y_pred = svm_model.predict(X_test)`

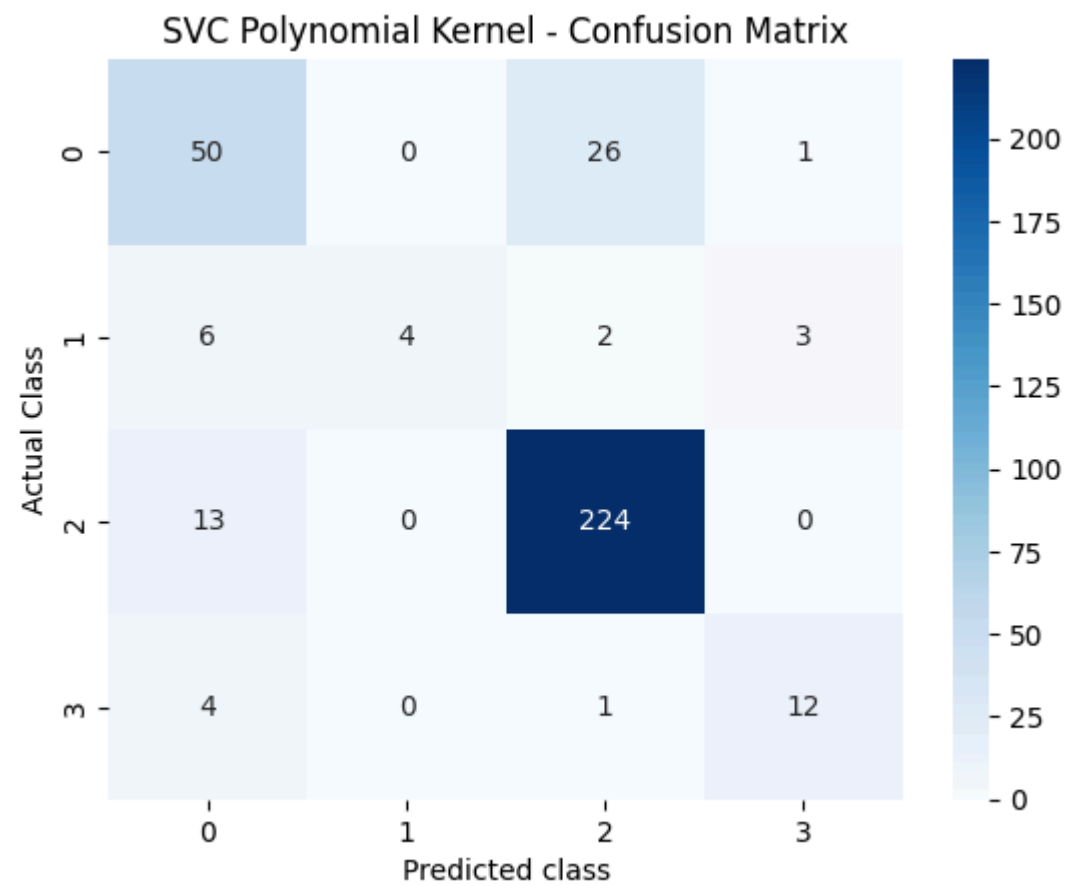
In [31]: `print("Accuracy:", accuracy_score(Y_test, Y_pred))`
`cm = confusion_matrix(Y_test, Y_pred)`
`print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))`

Accuracy: 0.838150289017341

Confusion Matrix:

```
[[ 50   0  26   1]
 [  6   4   2   3]
 [ 13   0 224   0]
 [  4   0   1  12]]
```

In [32]: `sns.heatmap(cm,annot=True,fmt="d",cmap="Blues")`
`plt.xlabel("Predicted class")`
`plt.ylabel("Actual Class")`
`plt.title("SVC Polynomial Kernel - Confusion Matrix")`
`plt.show()`



In []: